OBINNA DOMINIC UKE

Applied Environmental Economist ◇ Climate, Energy, and Finance

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Africa Business School, Mohammed VI Polytechnic University (UM6P), Rabat, Morocco
COPPE, Federal University of Rio de Janeiro (UFRJ), Rio de Janeiro, Brazil

FIELDS

Primary: Environmental and Energy Economics
Secondary: Financial Economics, Development Economics, Applied Econometrics

Research interests: Carbon pricing and fossil fuel subsidies; fossil revenue, carbon lock-in and transition risk; carbon border adjustment and carbon accounting; climate risk and financial markets; sovereign risk and credit markets; climate finance and adaptation in the Global South.

EDUCATION

Ph.D. in Management and Organizational Sciences 2023 to Aug. 2027 (expected)

Research field: Environmental and Energy Economics
Africa Business School, Mohammed VI Polytechnic University (UM6P), Rabat, Morocco

- Dissertation: *Climate Risk, Carbon Pricing, and Financial Markets: Essays in Environmental and Financial Economics*
- Supervisors: Prof. Amine Ben Amar (Finance Cluster, ABS, UM6P) and Prof. Olga Gorelkina (Moroccan Center for Game Theory, UM6P)

Joint Doctorate, Production Engineering (PEP/COPPE) 2023 to Aug. 2027 (expected)

Research field: Environmental and Energy Economics
COPPE, Federal University of Rio de Janeiro (UFRJ), Rio de Janeiro, Brazil

- Joint doctoral agreement with ABS, UM6P, conferring the doctoral degrees of both institutions on a single defence; co-supervisor Prof. Luan Santos

M.Sc. in Fertilizer Science and Technology Oct. 2020 to Jul. 2022
ESAFE, Mohammed VI Polytechnic University (UM6P), Benguerir, Morocco

B.Sc. in Agriculture (Soil and Crop Sciences) Dec. 2013 to Aug. 2018
Usmanu Danfodiyo University, Sokoto, Nigeria

JOB MARKET PAPER

"The Carbon Footprint of Fossil Windfalls"

Sole authored. October 2026. JEL: Q32, Q54, E62, F41, H54, O13.

Every rise in the world price of oil, gas or coal transfers income to producer economies, and the energy transition will make these transfers more volatile before it makes them smaller. Using national accounts, government revenue and sector-level fossil CO_2 for 163 countries over 1981 to 2024, and a windfall shock built from predetermined fossil rents and unexpected world fuel prices, I estimate what windfalls do to producer economies and their emissions. In the 75 economies with resource-revenue data, almost all of the windfall reaches the government budget; most is saved abroad in the first year and absorbed into consumption, investment and imports within two. Each dollar of windfall revenue adds 0.52 kg of CO_2 at home (95% weak-instrument-robust interval: 0.20 to 0.89 kg), about half the carbon intensity of an average dollar of GDP, mostly through road transport, power and buildings, with cement tracing an investment channel through existing kilns. Instrumenting the windfall with OPEC oil supply news reproduces the macroeconomic, construction, buildings and road transport responses, and an equally large rise in energy CO_2 in independent statistics. In cement, buildings and power, booms raise emissions and busts of the same size do not lower them. Over the 2003 to

2014 boom, windfalls added about 2.7 billion tonnes of CO_2, worth roughly 500 billion dollars at the social cost of carbon.

WORKING PAPERS

"Riding the Climate Wave: A Wavelet Analysis of Sovereign Risk in Developed and Developing Economies"

With Amine Ben Amar and Olga Gorelkina. Under review: International Review of Financial Analysis (FINANA-D-25-05094); revised version submitted September 2026. Presented: 2nd QFDAM, Amiens, June 2025; 71st EWGCFM, Rabat, May 2025.

Which dimension of climate risk do sovereign credit markets price, physical exposure or the capacity to adapt? Across 59 economies over 2008 to 2021, wavelet coherence shows co-movement between climate vulnerability and sovereign CDS spreads at short, medium and low frequencies alike, neither concentrated in the long horizon nor systematically different between advanced economies and EMDEs. In the cross-section the vulnerability gradient vanishes once development is controlled, while readiness carries a strong negative association with spreads net of income, public debt and governance: for EMDEs, closing the readiness gap is associated with roughly 74% smaller spreads. It is adaptive capacity, not exposure, that the market rewards. Significance is assessed against an AR(1) red-noise null with 10,000 Monte Carlo simulations. We also correct a measurement convention that inflates conventional coherence summaries.

"Fiscal Capacity, Not Climate Exposure: Decomposing Global Fossil-Fuel Underpricing and Its Equity Burden"

With Olga Gorelkina and Amine Ben Amar. Submitted: Energy Policy. JEL: Q48, Q54, Q58, H23, H20, O23.

We build a near-global panel of effective carbon prices for 174 countries from 2015 to 2022, integrating explicit instruments with implicit fossil fuel subsidies against three social cost of carbon benchmarks, and split the shortfall into a domestic wedge, made up of supply-cost gaps, foregone consumption taxes and local air pollution, and a global climate wedge. Only Sweden, Norway and Finland price above zero. Fiscal capacity, measured by non-resource tax revenue, predicts the domestic wedge, roughly two thirds of the total, while no observable state characteristic predicts the climate wedge. Vulnerability does not raise deficits directly, yet vulnerability and weak fiscal capacity coincide so closely that the income-normalised burden is about twenty times larger in the most vulnerable quintile. Domestic revenue mobilisation is therefore climate policy for the larger part of the shortfall; the climate wedge requires international coordination.

"Financial Reach and Household Coping under Climate Shocks: Evidence from Ten African Countries"

With Todd Cort (Yale School of Management). Submitted: World Development (WD-45053). JEL: Q54, O16, G51, D14.

Harmonised shock-and-coping modules from ten African countries permit a within-household comparison of responses to a covariate climate shock and to an idiosyncratic one, so household-fixed characteristics difference out. Covariate shocks are associated with a 1.9 to 2.2 percentage-point higher probability of reducing consumption, while idiosyncratic shocks more often trigger asset sales. In the 2018 West African sample a sender elsewhere in the country is associated with a 4.6 percentage-point smaller shift towards cutting consumption than a same-region sender, surviving region fixed effects, while credit access is indistinguishable from zero. Growing-season rainfall anomalies remain correlated across roughly 254 km on a CHIRPS correlogram, which motivates but does not identify the reach measure. The estimates are associational, and the contribution is conceptual as much as empirical: the exposure of the income backing a financial claim matters alongside the instrument a household holds.

"Fossil Fuel Underpricing and the Carbon Composition of Manufacturing Value Chains in Climate-Disaster Years"

Sole authored. Submitted: Economic Analysis and Policy. Replication package: 10.5281/zenodo.20468571. JEL: Q54, Q58, F18, L60, O14, H23.

Holding domestic energy prices below their efficient level can shield energy users from cost shocks, yet little is known about whether underpricing shapes how industrial structure adjusts to climate disasters. Combining global value chain output for 77 countries and 18 manufacturing sectors over 2016 to 2022 with IMF underpricing estimates and EM-DAT disaster records, I find that in disaster years the carbon-intensive share of manufacturing value-chain output rises with underpricing ($p = 0.006$), by about one percentage point at the 90th percentile, while it falls slightly and insignificantly where fuel is priced close to cost. The pattern holds with country-specific trends and a control for fossil rent exposure, and sector-level estimates with country-sector, country-year and sector-year fixed effects point the same way ($p = 0.041$). Sanctions and armed conflict show nothing comparable, and the association is strongest in resource-rich economies with weak democratic accountability. Because underpricing varies mainly across countries, the estimates compare pricing regimes and are conditional associations.

"Heat Beyond the Farm Gate: Offsetting Supply and Demand Responses in European Food and Drink Manufacturing"

With Amine Ben Amar and Olga Gorelkina. October 2026. JEL: C23, L66, Q11, Q13, Q54.

Heat reaches food and drink manufacturers through the supply of farm inputs and the demand for their products, and the two channels can move output in opposite directions. Using panel local projections identified from differences between countries' weather in the same month, we estimate monthly weather responses for food and drink industries in 21 European countries, and for raw milk deliveries and slaughter, over 2000 to 2026. In the average country a summer month 1°C warmer than normal lowers raw milk deliveries by 0.3% and bakery output by 0.8% and raises beverage output by 2.0%. The milk and meat losses follow days above 30°C, occur in warm but not cool months and are not recovered, as livestock heat stress would produce; beverage output follows both kinds of heat and moves with the price of non-alcoholic beverages, as a demand shift would. Because the responses offset, 81% of the gross change in value added cancels and beverages are the only one of 24 manufacturing divisions to respond. Aggregate food output therefore understates how exposed the livestock-based food chain is to heat.

"Building Resilience, Reducing Risk: A Systematic Review of Climate Resilience, Climate Finance Impacts, and Market Mispricing in the Global South"

With Amine Ben Amar and Olga Gorelkina. Retained as dissertation material.

A PRISMA-based review of the relationship between climate resilience initiatives, financial risk reduction and market mispricing in the Global South. Firms in climate-vulnerable regions face higher borrowing costs and stricter loan terms, with governance and precautionary financial policies the critical moderators of climate finance impact.

WORK IN PROGRESS

"Carbon Accounting Capability and Exposure to the EU Carbon Border Adjustment Mechanism: Evidence from African Exporters"

With Luan Santos (COPPE, UFRJ). Target: Business Strategy and the Environment.

Border carbon pricing is usually studied as trade policy falling on countries. It falls on consignments, and charges a conservative default unless a producer verifies actual emissions inside the regulatory boundary. The capability it rewards is boundary fit: emissions measured on the system boundary the regulating jurisdiction prices. Computing liability line by line for 1,760 eight-digit tariff codes from European import declarations and official default values, African exporters face EUR 1,768.1 million annually on the 2023 to 2025 mean at full phase-in, with incidence reaching 49.0% of customs value on semi-finished steel against a zero applied tariff, so the mechanism reintroduces the tariff escalation that trade policy removed. Five installation cases show boundary fit distributed independently of ownership form, firm size and country income: the best-placed producer is 94% state-owned, while a listed multinational's subsidiary publishes nothing at installation level.

"Climate Shocks and Food Manufacturing: Aggregation Bias in the United Kingdom"

With Amine Ben Amar and Olga Gorelkina. Presented: 3rd QFDAM, Rabat, June 2026.

Climate-macro studies typically model the food system as a single sector. Using UK quarterly data for 1990 to 2025, we separate primary agriculture from downstream food manufacturing and ask what a single food-sector climate coefficient conceals. A food-system VARX estimated on the same data has an impact response less than half the agricultural one, and a Monte Carlo experiment shows it systematically erases the persistent decline the disaggregated estimates imply, so the coefficient depends on sectoral composition and is not a structural parameter. Sectoral estimates are being rebuilt.

"Sovereign Climate Risk and Corporate Bond Spreads: Evidence from Emerging Markets"

With Luan Santos (COPPE, UFRJ).

Monthly panel of emerging market corporate bonds from 2018 to 2024, matched to sovereign CDS, rating actions, EM-DAT disasters and measured tropical cyclone tracks, with pre-registered hypotheses and randomisation inference. Current estimates find no sovereign transmission, no change in market access and no average corporate repricing after disasters, and do not support state backing as an insulation mechanism. Developed under the joint doctoral agreement with COPPE, UFRJ.

PEER-REVIEWED PUBLICATIONS

Guéablé, Y.K.D., **Uke, O.D.** et al. (2022). Study of the Sodicity of Phosphate By-Products and Sludge Mixture for Large-Scale Application in Mine Site Reclamation. *Environmental Sciences Proceedings*, 16(1), 18.

Uke, O.D. & Haliru, M. (2021). Salinity Study of the Soils of Fadama Farms, Sokoto, Nigeria. *Farming and Management*, 6, 1-7.

Yusuf, Y.M., Bagudo, H.A. & **Uke, O.D.** (2021). Effects of Weed Control on Growth and Yield of Pearl Millet in the Sudan Savanna, Sokoto State, Nigeria. *Journal of Current Opinion in Crop Science*, 2(2), 155-164.

CONFERENCE PRESENTATIONS

3rd Quantitative Finance and Data Analytics Meeting (QFDAM 2026), Rabat, Morocco June 2026
Climate Shocks, Sectoral Transmission, and Monetary Policy: Evidence from UK Food Manufacturing

2nd Quantitative Finance and Data Analytics Meeting (QFDAM 2025), Amiens, France June 2025
Riding the Climate Wave: Wavelet Analysis of Sovereign Risk in Developed and Developing Economies

71st EURO Working Group for Commodities and Financial Modelling (EWGCFM 2025), Rabat, Morocco May 2025
Riding the Climate Wave: Wavelet Analysis of Sovereign Risk in Developed and Developing Economies

TEACHING

Teaching Assistant, Financial Econometrics [Master 2] Sep. to Nov. 2026
Mohammed VI Polytechnic University (UM6P) (Instructor: Prof. Amine Ben Amar)

Teaching Assistant, Microeconomics [Licence 1] Sep. to Nov. 2026
Mohammed VI Polytechnic University (UM6P) (Instructor: Prof. Amine Ben Amar)

Teaching Assistant, Essentials of Financial Markets [M1] Jan. to Feb. 2025
Africa Business School, UM6P (Instructor: Prof. Amine Ben Amar)

PROFESSIONAL EXPERIENCE

Regional Field Project Manager Jul. 2022 to Oct. 2023
OCP Africa Nigeria

Coordinated multi-regional agricultural development programmes across Southeast and South-South Nigeria in partnership with USAID/TradeHub, managing budgets, donor reporting and workplans, and produced policy briefs on climate-smart agriculture for government counterparts.

Research Agronomist / Blending Plant Support Jun. 2021 to Jun. 2022
OCP Africa Nigeria

Designed and executed fertilizer validation field trials; prepared grant applications and supported monitoring and evaluation for USAID-funded agricultural initiatives.

Founder and Chief Technical Officer Dec. 2020 to Jan. 2022
AgriBox SARL, Morocco

Founded a climate-smart hydroponic enterprise; developed impact investment proposals and private sector partnerships.

Project Manager, Microfinance Operations Jun. 2019 to Mar. 2020
Bincom ICT Solutions (Social Lender), Nigeria

Managed microfinance loan disbursement, risk assessment and financial proposals across Kano State, and served as institutional liaison between Social Lender and the banking sector.

FELLOWSHIPS AND AWARDS

Full Doctoral Merit Scholarship, Africa Business School, UM6P, Morocco	2023
HULT Prize in Innovation, Mohammed VI Polytechnic University	2021
NARSAD Deliberative Polling Award, UM6P and Stanford University	2021
IBN Rochid Foundation Scholarship, M.Sc. Studies, OCP Group	2020

TECHNICAL SKILLS AND LANGUAGES

Statistical software: R, Stata, Python, SQL, GIS, VS Code, Git, Advanced Excel.

Methods: shift-share instruments and local projections, including time-varying local projections with stochastic volatility; SVAR with recursive identification; wavelet and partial wavelet coherence; saturated panel fixed effects with cluster-robust inference; within-household first differences; IV and weak-instrument diagnostics; randomisation inference; PRISMA review.

Data and geospatial: construction and validation of multi-source country, sector, firm and household panels; CHIRPS, CHIRTS, ERA5 and Sentinel-1; Eurostat Comext and UN Comtrade at tariff-line level; World Bank LSMS-ISA and EHCVM surveys; EDGAR sector emissions; IMF fossil fuel subsidy and carbon pricing data; reproducible pipelines with replication packages on Zenodo.

Languages: English (native), Igbo (native), Portuguese (intermediate), French (basic).